Introduction

A photovoltaic (PV) cell is a device capable of converting light power into electric power. Such devices are characterized by a high adsorption index in the IR/VIS range, where sun's spectral power is maximum. Semiconductors are ideal for this kind of application, because of their high absorption of photons with energy $\hbar\omega > E_g$, where E_g is the energy forbidden gap. In this class of materials, adsorbed photons generate electron-hole pairs, which quickly lose energy interacting with lattice; until the electron reaches bottom of the conduction band and the hole top of the valence band. After that, the pair can recombine radiatively (producing a photon with energy equal to the energy gap) or non-radiatively. In order to collect free carriers, the pair has to be separated before recombination, for example generating an electric field inside the device. The pair can be also bounded due to Coulomb interaction, forming a hydrogen-like bound state called exciton. In this situation, the energy needed to separate the two carriers increases by the binding energy of the exciton.

Photovoltaic Characteristics

Solar cells are actually a p-n junctions with rectifying behaviour, pairs produced by the adsorption of a photon are accelerated by the electric field inside the depletion region of the diode. For this reason a short-circuit current I_{sc} will flow even in the absence of any applied voltage. The I-V characteristics is:

$$I = I_{sc} - I_s \left[\exp\left(\frac{eV}{\eta k_B T} \right) - 1 \right], \qquad (1)$$

where η is the ideality factor of the diode. Another characteristic quantity is the open circuit voltage V_{oc}, defined as the voltage inside the device when $I = 0$. With reference to Fig. (1), we define the fill factor (FF) of the diode as the ratio between the maximum power the cell can generate and the product $V_{oc} I_{sc}$.

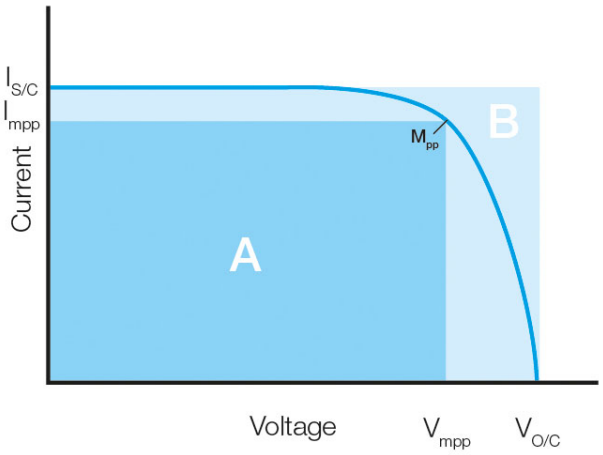


Figure 1: The fill factor is defined as the ratio between the area A and the area A+B.

Sun has an irradiance spectrum quite similar to a black-body spectrum with temperature $T_s \simeq 5700K$. The atmosphere is responsible for a reduction of the irradiance, primarily near the adsorption lines of the air molecules (O_2, H_2O and CO_2). Optical path length of Sun's light depends on latitude, therefore in laboratory experiments is used a universal spectral irradiance named AM 1.5 (equivalent Air Mass), which corresponds to mid latitudes (48.2°). The corresponding integral irradiance is 930 W/m^2. Solar cells are able to convert only a tiny fraction of this energy: technology progress is quantified by the increase of the power conversion efficiency (PCE) of solar cell.

Schockley-Queisser Limit

Before the commercialisation of the first solar cell, scientists debated about the maximum PCE that a single cell can reach. Semi-empirical calculations [7] showed that the upper limit was about 22%; adsorption and emission depend on a large variety of phenomena, therefore the calculation of a theoretical limit requires such a great effort. The simplest limit is the Carnot limit: the efficiency of a perfect engine which works between T_s and $T_c = 300$ K is $1 - T_c/T_s \simeq 0.95$. The real limit for a semiconductor junction is actually much lower: photons with energy lower than the bandgap are not adsorbed and the energy difference between the photon energy and E_g is lost. Schockley and Queisser [16] found an E_g dependent PCE limit, starting from these assumption and making simple thermodynamical considerations. In a junction with infinite electron-hole mobility, at 0 K, with unitary adsorption index over the energy gap, with lossless geometry; the PCE is given by NE_g/E_s, where N is the number of photons over E_g:

$$N = \int_{E_g}^{\infty} dE\ g(E)\ (\exp(\beta_s E) - 1)^{-1}, \qquad (2)$$

according to Bose-Einstein statistic. The solar energy is:

$$E_s = \int_{0}^{\infty} dE\ g(E)\ E\ (\exp(\beta_s E) - 1)^{-1}. \quad (3)$$

The maximum PCE results in 44% at $E_g = 1.1$ eV; it's important to notice that Si cells should display one of the greatest PCE, having $E_g = 1.1$ eV. At room temperatures, however, the entire cell emits as a black body; because of the high absorption index. In semiconductor physics this behaviour is translated to the fact that it's impossible to completely avoid radiative recombination, technology improvements are related to the suppression of the non-radiative ones. The final SQ limit is about 33% (AM 1.5), for $E_g = 1.3$ eV.

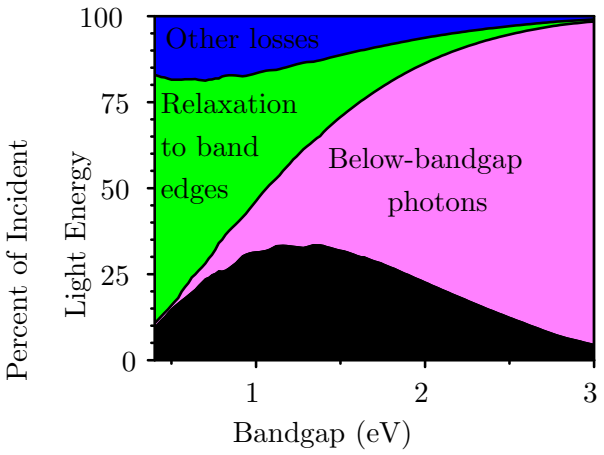


Figure 2: All contributes to SQ limit: the black curve is the effective PCE, the blue strip derives primarily from recombination processes.

Why Perovskites?

Silicon is still the most used semiconductor; commercial panels have reached PCE of 20%, while research records are 26.7% for mono-Si and 23.3% for multi-Si [1]. Such panels are relatively cheap, durable (>60% efficiency after 25 years [2]) and with very high efficiency. On the other hand, perovskites have reached desruptive results, with a PCE of 25.7% in about ten years of research. The biggest con of this emerging technology is the extremely low cost: solution growth processes are less expensive and less energy consuming than Si single crystal growth. The major problems of perovskites are the insta-

bility in normal operation conditions and toxicity: the most studied perovskites degrade within days into highly toxic Pb-halides.

Structural and Optical properties of Perovskites

Perovskites are a family of materials with a specific crystal structure. Their chemical formula is ABX_3, with A representing a large organic/inorganic ion (such as methylammonium – MA^+, formamidinium – FA^+, Cs^+), B is a IVA element (Pb, Sn, Ge) and X is a halogen (like I, Br, Cl).

Perovskites are studied because of their astonishing optoelectronic proprieties [10], such as high adsorption, high mobility and diffusion length of free carriers and relatively low exciton binding energy. For this reason, macroscopic electrical properties of the cell, like I_{sc}, V_{oc}, FF and so the overall PCE are enhanced. The energy gap, the adsorption edge and stability can be tuned by regulating ion position or by replacing them. Absorption and photoluminescence (PL) experiments allow investigation of the energy gap, spectral adsorption and quantum yield for LED research. At low temperature, the energy peaks underline exciton binding energies. A large data is compared with theoretical and DFT calculations [13]. As an example on how microscopical properties affect PCE, $MAPbX_3$ perovskite show decreasing bandgap and exciton binding energy, increasing atomic number of halogen. This behaviour results in $MAPBI_3$ having the overall highest PCE with same conditions.

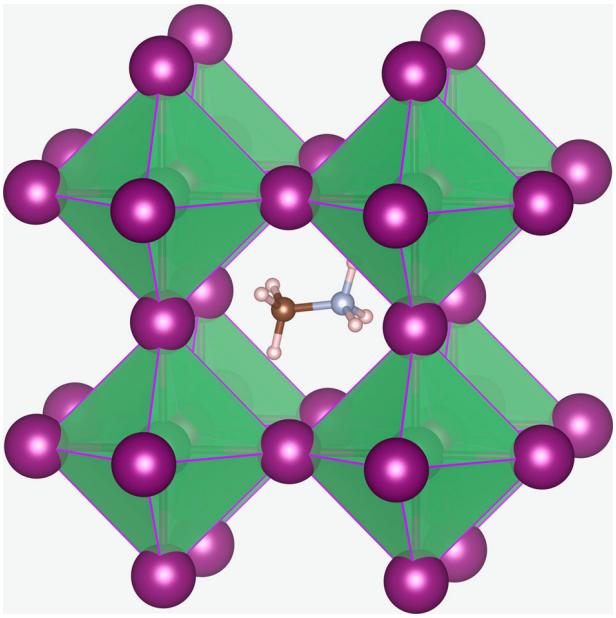


Figure 3: Crystal structure of $MAPbBr_3$. MA is at the center of the structure (simple cubic); Br (purple) forms an octahedral structure (face centered cubic); Pb is inside the octahedron (body centered cubic).

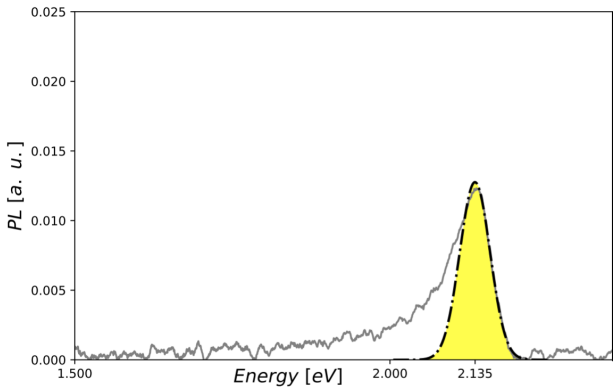


Figure 4: PL experiment on single crystal $MAPbBr_3$ at room temperature.

Stability and Toxicity

For the same perovskite, three crystal phases are known: hexagonal/tetragonal, orthorhombic/rhombohedral and cubic. Cubic structure displays an enhanced intrinsic stability and electrical properties. Empirical formulas [5] predict 92% of crystal structure, given the ionic radii of the three species and oxidation state of A species. Crystal phase can be tuned by annealing at a certain temperature [3]. Bulk perovskite has a surprising defect tolerance, due to structural stability and high formation energies of unwanted defects [11]. Doping can be done simply by varying the concentration of the species: Pb-poor/rich environment leads to p/n-type. However, heat, humidity, light and interface materials have a large effect on extrinsic stability, which leads to degradation of the surface layer. Moisture and other polar solvents are able hydrate the crystal, which degrades into Pb-halides; the degradation can be tracked measuring the intensity of halides XRD peaks. The degradation time can be increased by well encapsulating the active layer, by choosing a transport layer that prevents underneath perovskite from H_2O contact. Also high temperatures affect the durability of the cell: performance are slightly reduced under 24h illumination at 85°C for $MAPbI_3$ cell, under inert environment (O_2, N_2 rich). In this case, high concentration of Pb-halides is found (through XRD/XPS) only in the cell exposed to normal air.

Toxicity is related to the formation of Pb-halides escaping from the perovskite layer, because of the degradation of the cell. New classes of Pb-free materials are studied, however, $APbX_3$ still has the most promising performances. The $ASnI_3$ class has reached a PCE of 9.6%, these materials display and ideal bandgap,

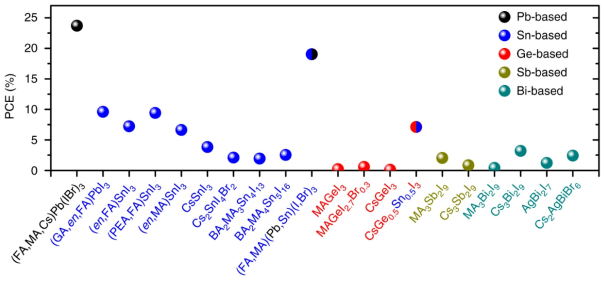


Figure 5: PCE of most popular lead-free perovskites compared to $APbX_3$ [9].

in the 0.9-1.6 eV region. Although microscopical properties are similar to the Pb counterpart, such devices have low V_{oc}, probably due to the heavy p-type, attributed to the oxidation of Sn^{2+} into Sn^{4+}. The substitution with Ge seems to little increase device stability, but because of the bigger ($\simeq$ 2 eV) energy gap this devices have poor results. DFT investigations are performed in the large class of double perovskites A_2BX_6 [14] [8], the aim is to find the best combination of elements and then the best theoretical PCE.

Cell Structure

A solar cell is not only composed of the semiconducting active layer: the most used setup is the p-i-n, where perovskite represents the intrinsic active layer. The two dopant layers are responsible for the generation of an electric field capable of separate the photoexcited pair. With reference to Fig. (6), the heterojunction formed in this way allows electron and hole one-way transport. For this reason, the n-doped layer is also called Electron Transport Layer (ETL) and the p-doped Hole Transport Layer (HTL). The two electrodes constitute the external layers. Naturally, at least one of the metal electrode has to

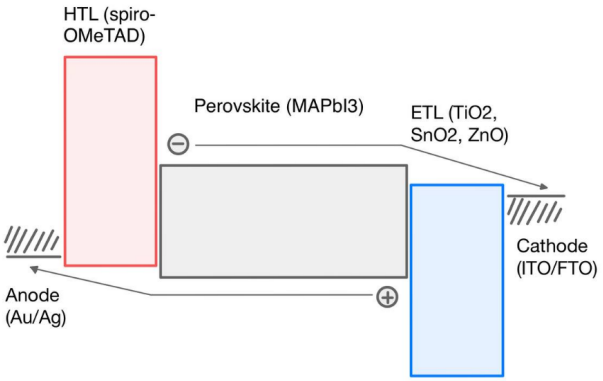


Figure 6: Schematic energy configuration of n-i-p junction. Electrons are conducted into cathode passing through the conduction band of ETL, holes pass through HTL. Under parenthesis are found the most used material for that scope.

be transparent (neither absorptive nor reflective) to the light frequencies that the active layer can absorb. The opaque electrode is typically Au or Ag, the transparent electrode is ITO/FTO (Indium Tin Oxide, Fluorine doped Tin Oxide).

Heterojunctions and Contacts

Materials used is solar cells have to be chosen properly: a slight change in the concentration of dopants, width of a layer, etc... results in a dramatic change in the PCE. The p-i-n perovskite cell, in the language of semiconductor physics, consists in two heterojunctions (formed between the active layer and the transport layers) and two metal-semiconductor contacts. The heterojunction must inhibit charge transport in one direction, allowing (or improving) the flow of the charge in the opposite direction. Since two materials in a heterojunction will have different energy bandgaps, a discontinuity of the valence and conduction band will be found at the inter-

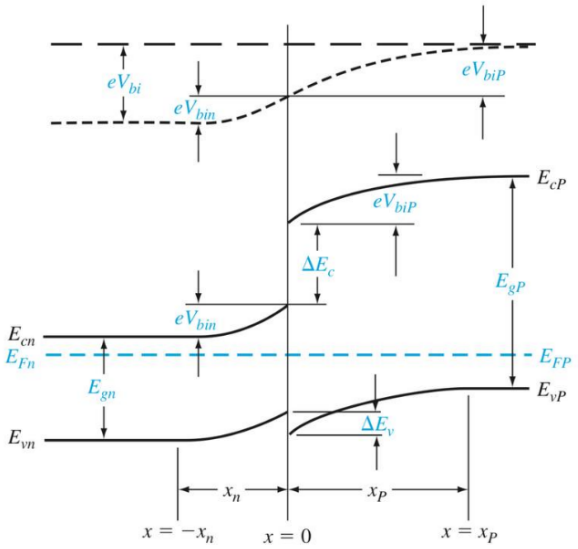


Figure 7: Energy-band diagram of a slightly n-doped and a highly P-doped heterojunction, in thermal equilibrium [12]. The local vacuum levels differs from the previous one by the built-in potential V_{bi}. In this junction, the conduction of electron from left to right is inhibited by the barrier $\Delta E_c + V_{bi}$.

face between these two materials. In this way, a built-in potential is generated, which block free charges in one direction. In order to obtain a high quality heterojunction, the lattice constants of the two materials must match, with a displacement <5%. Mismatch causes stress, the lattice relaxes itself generating dislocation defects. Every material has its own vacuum level, defined as the energy level where an electron is free and has zero kinetic energy. Once the position of valence, conduction and Fermi energy bands are known, the energy-band diagram (defined as the energy of these levels as a function of distance) can be drawn. The energy difference between the two Fermi levels is equal to $-eV$, where V is the voltage difference between the two species. The vacuum level can vary with position, but it must be continuous; and the electron affinity/work function (defined as the distance between conduction band/Fermi energy and vacuum level) must be constant.

Also electrode's material has to be chosen properly, in order to obtain an ohmic metal-semiconductor contact. An ohmic contact is a low resistance junction, with a ideally linear I-V characteristics. It is obtained when the band bending that occurs in the depletion region creates almost negligible discontinuity at the interface. This requirement is fulfilled when the work function of the metal is greater (lesser) than that of the semiconductor in case of n (p) doping. When an external voltage is applied, in an ideal contact, the bands of the semiconductor bend linearly (Fig. (8)).

Growth Methods

Thin film deposition is the technique referred to the growth of nm-μm multilayer structures.

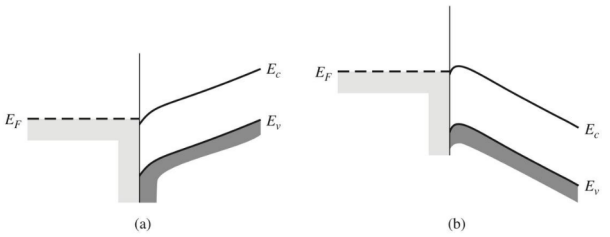


Figure 8: Ideal energy-band diagram of (a) a negatively biased semiconductor, (b) a positively biased one [12]

Perovskite, with respect to single crystal silicon, can crystallise from solution: first, a liquid layer of perovskite is deposited onto the substrate, and then a polysrystalline layer is formed, via anti-solvent techniques or by simply annealing. Temperatures involved in this process are crucial, because of the poor thermal stability of perovskites. A widely used technology regarding thin film deposition is certainly spin coating: with this technique the solution is simply dropped onto the substrate or the previous layer and then the liquid is spread via centrifugal force. This is one of the most used methods in laboratories, because it is simple, fast and cheap and solar cells with highest PCE are fabricated with this method. However, spin coating suffers from poor reproducibility and cannot be used for large areas for commercial purposes; furthermore, it is not easy to guarantee good thickness uniformity across the cell. For this reason a growing interest is shifting towards alternative scalable deposition techniques from solution, such as blade-coating, slot-die coating, inkjet printing and spray coating [15] [6].

On the other hand, perovskite single crystal cells exhibit much longer recombination time [17], due to lower trap state densities. This results in higher diffusion length and absorp-

tion, with overall high theoretical PCE compared to polycrystalline counterpart. Also, polycrystalline cells suffer from degradation at grain boundaries, avoided using single crystals. However, single crystal PCE lab records are far lower than its competitor, because of the actual impossibility of growing a crystal of uniform thickness.

Tandem Cells

The SQ limit is naturally referred to single cell. If a device is composed of more than one active layers, with semiconductors of different bandgap, the theoretical limit increases. A tandem cell is made of two layers, in which the outer semiconductor has higher bandgap and therefore efficienlty aborbs high energy photons, without losing energy due to band relaxation. Low energy photons are transferred and absorbed by the second layer (Fig. (9)). Fig. (10) displays the new SQ limit for a tandem cell: the maximum is for 0.96/1.63 eV bandgaps and corresponds to a 47% PCE. Increasing the number of layers, also the PCE increases, reaching 60% with 6 layers.

Also in this field of research, perovskite has a strong influence: Si/perovskite tandem are reaching a 30% PCE, exploiting the 1.1 Si/1.68 $Cs_{0.05}(FA_{0.77}MA_{0.23})_{0.95}Pb(I_{0.77}Br_{0.23})_3$ bandgaps [4].

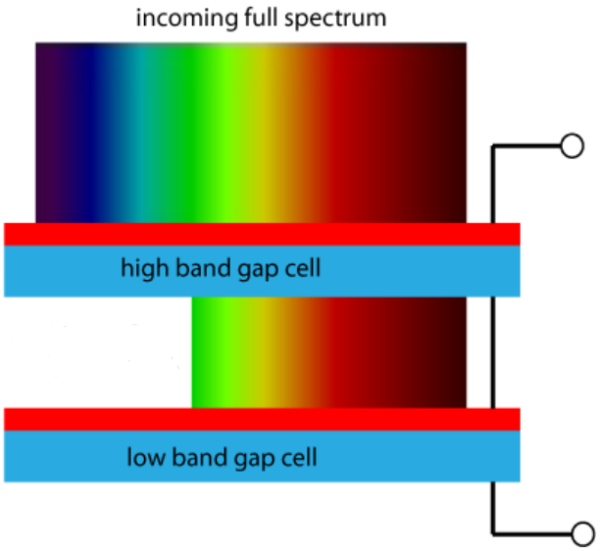


Figure 9: Schematic picture of tandem solar cell. The exterior (high bandgap) active layer absorbs high frequency radiation and transmit lower frequency photons to the interior layer, with lower bandgap.

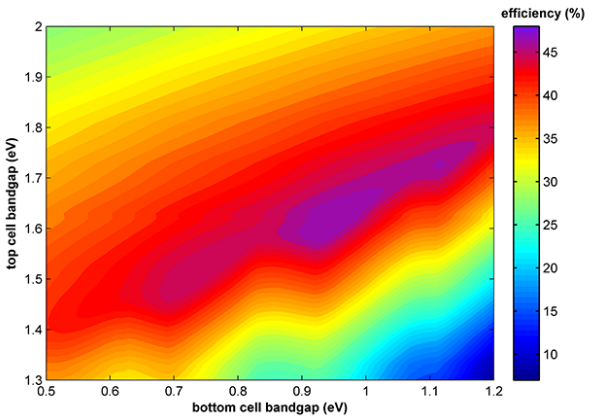


Figure 10: Maximum theoretical PCE obtained with two different bandgap semiconductors.

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